The Absorber and the Window: Wheeler–Feynman Theory and the Temporal Propagator

Light-Meter Physics

Abstract

Wheeler–Feynman absorber theory gives a time-symmetric, direct-action account of electromagnetic interaction in which the response of other charges is invoked to explain radiation reaction and the observed retarded radiation description. This paper introduces that proposal and its absorber assumptions, then derives a separate mathematical construction: finite averages of a signal over future- and past-sampling time windows. The window operators admit an exact even–odd decomposition and, under suitable regularity, a derivative expansion. The shared use of symmetric and antisymmetric parts is a structural analogy only. Finite time-window averages are not electromagnetic Green functions and do not reproduce the absorber dynamics. A possible weak-field relation between an acceleration average and a clock-rate comparison is stated as an additional model step, with the required path and time-coordinate assumptions made explicit. The aim is an accessible comparison that preserves the distinction between physical theory, exact operator algebra, and project-specific modeling.

Keywords: Wheeler–Feynman absorber theory, direct action, advanced and retarded Green functions, radiation reaction, finite-window average, temporal propagator

1 Introduction

Electromagnetic radiation presents an apparent asymmetry. The fundamental field equations admit both retarded solutions, supported by sources in the past light cone, and advanced solutions, supported by sources in the future light cone. Ordinary observations are described using retarded radiation. Wheeler and Feynman asked whether this familiar direction could be understood from a time-symmetric interaction between charges together with the response of the rest of the charged universe, rather than imposed as a separate radiation condition [1, 2].

This paper explains that absorber proposal and a mathematical object called the *temporal propagator* in the Light-Meter Physics materials. Here the latter term means a finite interval average of a specified time-dependent signal. It is not a standard electromagnetic Green function. Both topics can be organized into symmetric and antisymmetric parts, but they act on different objects and answer different questions. The absorber theory concerns interactions among charges and the boundary conditions on their collective response; the finite-window operators are exact averages defined directly from a function of time.

The distinction matters because similar algebra does not establish a physical mechanism. In particular, the two-window identities below do not derive radiation reaction, impose a cosmological absorber condition, or show that advanced electromagnetic fields are finite-duration averages. A possible connection to gravitational clock-rate comparisons is also conditional on a specified field, path, and time mapping; it is not a consequence of Wheeler–Feynman electrodynamics.

2 Retarded and advanced solutions

For a linear wave equation, retarded and advanced Green functions are distinct solutions with different support conditions. Schematically, if $\mathcal{L}$ denotes the relevant wave operator and x, x' spacetime events, they satisfy

$$\mathcal{L}G_{\mathrm{ret}}(x,x') = \delta(x-x'), \qquad \mathcal{L}G_{\mathrm{adv}}(x,x') = \delta(x-x'), \tag{1}$$

with retarded or advanced boundary conditions, respectively. The normalization and tensor structure depend on the field and convention; Eq. (1) displays only the shared schematic source equation.

The time-symmetric and antisymmetric combinations can be defined as

$$G_{\mathrm{sym}} \equiv \frac{1}{2}\left(G_{\mathrm{ret}} + G_{\mathrm{adv}}\right), \qquad G_{\mathrm{rad}} \equiv \frac{1}{2}\left(G_{\mathrm{ret}} - G_{\mathrm{adv}}\right). \tag{2}$$

Thus $G_{\mathrm{ret}} = G_{\mathrm{sym}} + G_{\mathrm{rad}}$ and $G_{\mathrm{adv}} = G_{\mathrm{sym}} - G_{\mathrm{rad}}$. These expressions concern spacetime Green functions: their support lies on or within the appropriate light cone, depending on the equation and spacetime.

In Wheeler–Feynman electrodynamics, the interaction is formulated directly between charged particles using a time-symmetric kernel, rather than treating the electromagnetic field as an independent intermediary with its own freely specified initial data. The Green-function language is a useful way to display the time symmetry of the interaction; it should not be

confused with an average of a scalar signal over a chosen time interval. The 1949 paper develops the direct-interparticle-action formulation, following the absorber proposal of 1945 [1, 2].

3 The absorber proposal and radiation reaction

The absorber argument considers the source and the other charged particles as a coupled system. In the time-symmetric description, the source's interaction has equal retarded and advanced parts. The source perturbs absorber charges; their collective response then acts back on the source. Under the theory's idealized absorber conditions, that response supplies the radiative contribution needed for radiation damping, while the field inferred at distant points has the familiar retarded form.

The familiar half-sum notation gives a compact picture of the proposal:

$$F^{\rm sym}_{\rm source} = \frac{1}{2}\left(F_{\rm ret} + F_{\rm adv}\right), \qquad F_{\rm eff} \overset{\text{ideal absorber}}{\simeq} F^{\rm sym}_{\rm source} + \frac{1}{2}\left(F_{\rm ret} - F_{\rm adv}\right) = F_{\rm ret}. \tag{3}$$

Equation (3) is schematic bookkeeping for the idealized source-plus-absorber argument, not a stand-alone identity for arbitrary electromagnetic fields. In the theory, the second term represents a collective absorber response under specified conditions; it is not generated by taking a time average of the source signal. The same response is invoked in the account of radiation reaction on the emitter.

An advanced solution is a mathematically permitted field configuration whose source lies to the future of the observation event. Its appearance in a time-symmetric formulation does not, by itself, describe a controllable signal sent into the past. The absorber story concerns correlations and boundary conditions across the interacting system, not an experimental communication protocol using advanced waves.

4 What the absorber condition assumes

Time symmetry alone does not select retarded rather than advanced radiation. The absorber argument requires additional physical input about the absorber's response and the relevant boundary or statistical conditions. In the classical proposal, the idealization of complete absorption is central to obtaining the desired collective response [1]. Whether a given cosmological model meets a complete-absorption condition is a separate problem, not a consequence of the local half-retarded plus half-advanced interaction. Davies analyzed this

cosmological requirement for classes of models, illustrating that the condition must be checked rather than assumed to hold universally [3].

Accordingly, this paper presents Wheeler–Feynman theory as a physical proposal with explicit conditions, not as a derivation of the radiation arrow from time symmetry alone or as a settled description that follows for every possible universe. Later quantum direct-action developments are beyond the present scope. The restricted goal is to explain the classical absorber mechanism at a conceptual level and to keep its physical assumptions visible.

5 The temporal propagator as a finite-window average

Now consider a distinct mathematical construction. Fix a positive interval $\tau_L = L/c$, a specified time coordinate t, and an integrable scalar signal $g(t)$. Define the forward- and past-sampling averages by

$$\mathcal{P}_+[g](t) \equiv \frac{1}{\tau_L}\int_0^{\tau_L} g(t+u)\,\mathrm{d}u, \qquad \mathcal{P}_-[g](t) \equiv \frac{1}{\tau_L}\int_0^{\tau_L} g(t-u)\,\mathrm{d}u. \tag{4}$$

Both operators preserve a constant input. They average one prescribed function over two intervals of equal duration; neither includes spatial propagation, a source-to-observer light cone, or a response of absorber particles.

Let $D \equiv \mathrm{d}/\mathrm{d}t$. Translation of the input gives the compact notation

$$\mathcal{P}_+ = \frac{e^{\tau_L D}-1}{\tau_L D}, \qquad \mathcal{P}_- = \frac{1-e^{-\tau_L D}}{\tau_L D}. \tag{5}$$

The apparent singularities at $D = 0$ are removable: the scalar functions $(e^x-1)/x$ and $(1-e^{-x})/x$ have finite limits there. Equation (5) is shorthand for the exact translation averages in Eq. (4); it does not require an inverse-derivative operation or an integration constant. In particular, $\mathcal{P}_-$ is not the inverse of $\mathcal{P}_+$. The identity $\mathcal{P}_- = e^{-\tau_L D}\mathcal{P}_+$ simply relates the two translated windows.

If g has a Taylor expansion that converges appropriately throughout each averaging interval, integrating term by term yields

$$\mathcal{P}_\pm[g](t) = \sum_{n=0}^{\infty} \frac{(\pm\tau_L)^n}{(n+1)!} D^n g(t) = g \pm \frac{\tau_L}{2} j + \frac{\tau_L^2}{6} s \pm \frac{\tau_L^3}{24} c_r + \cdots, \tag{6}$$

where $j \equiv Dg$ is jerk, $s \equiv D^2 g$ is snap, and $c_r \equiv D^3 g$ is crackle. Every derivative is evaluated at t and is a derivative of the specified signal in the specified coordinate time. For a merely integrable signal, the integral averages remain defined even when the infinite derivative series

is not. A finite Taylor truncation should be accompanied by its remainder when used as an approximation.

For a linear ramp $g(t+u) = g(t) + ju$, direct integration gives $\mathcal{P}_\pm[g](t) = g(t) \pm j\tau_L/2$ exactly. This simple check shows how the window direction appears in the odd derivative term without invoking any electromagnetic absorber.

6 Even and odd parts of the window pair

Adding and subtracting the window operators separates parity. From the integral definitions,

$$\frac{\mathcal{P}_+[g](t) + \mathcal{P}_-[g](t)}{2} = \frac{1}{2\tau_L}\int_{-\tau_L}^{\tau_L} g(t+u)\,\mathrm{d}u, \tag{7}$$

$$\frac{\mathcal{P}_+[g](t) - \mathcal{P}_-[g](t)}{2} = \frac{1}{2\tau_L}\int_{0}^{\tau_L} \left[g(t+u) - g(t-u)\right]\,\mathrm{d}u. \tag{8}$$

The first is a centered average over total width $2\tau_L$; the second compares the two half-windows with opposite signs. In operator notation, with $x \equiv \tau_L D$,

$$\frac{\mathcal{P}_+ + \mathcal{P}_-}{2} = \frac{\sinh x}{x}, \qquad \frac{\mathcal{P}_+ - \mathcal{P}_-}{2} = \frac{\cosh x - 1}{x}. \tag{9}$$

The even part retains even derivatives of g; the odd part retains odd derivatives. Their first terms are

$$\frac{\mathcal{P}_+[g] + \mathcal{P}_-[g]}{2} = g + \frac{\tau_L^2}{6}D^2 g + \frac{\tau_L^4}{120}D^4 g + \cdots, \tag{10}$$

$$\frac{\mathcal{P}_+[g] - \mathcal{P}_-[g]}{2} = \frac{\tau_L}{2}Dg + \frac{\tau_L^3}{24}D^3 g + \cdots. \tag{11}$$

The hyperbolic notation is a compact form of the exact integral relations; its derivative series has the same convergence qualifications as Eq. (6).

The window pair also has a simple frequency response. For the convention $g(t) = e^{i\omega t}$, direct integration gives

$$\mathcal{P}_\pm(i\omega) = e^{\pm i\omega\tau_L/2}\,\mathrm{sinc}\!\left(\frac{\omega\tau_L}{2}\right), \qquad \mathrm{sinc}(y) \equiv \frac{\sin y}{y}. \tag{12}$$

The sinc factor is the averaging envelope; the phase records that the forward or past window is centered half an interval away from t. This is the transfer function of a finite-window average. It is not the frequency-domain retarded or advanced electromagnetic Green function.

7 A structural analogy, not an identification

The two constructions can be compared at the level of their paired decompositions:

Feature	Wheeler–Feynman theory	Temporal-window operators
Mathematical object	Electromagnetic interaction kernel	Average of a specified function $g(t)$
Pair being split	Retarded and advanced Green functions	Future- and past-sampling windows
Physical ingredients	Charges and absorber response	Interval length and signal history
Role of the split	Describes time-symmetric interaction and absorber response	Separates even and odd signal contributions

The common feature is the organization into symmetric and antisymmetric parts. The physical objects, inputs, and consequences are different. Green functions encode spacetime support and the field equation; $\mathcal{P}_\pm$ act on a one-dimensional time history over finite intervals. The window algebra neither supplies the absorber's advanced response nor proves the complete-absorber condition. Conversely, Wheeler–Feynman theory does not imply the finite-window derivative ladder or its sinc response.

For this reason, the terms *retarded* and *advanced* should be reserved for Green functions and source support when discussing electrodynamics. When describing $\mathcal{P}_\pm$, *past-sampling* and *future-sampling* are less ambiguous: the labels describe which values of g enter the integral, not causal boundary conditions for a field equation.

8 A conditional interface with clock-rate models

Some Light-Meter Physics materials use a temporal-density convention $\delta\tau \equiv \mathrm{d}t/\mathrm{d}\tau$, where t is a specified static coordinate time and τ is the proper time of a static observer. This is a project-specific reciprocal clock-rate variable. In a static spacetime, the gravitational frequency shift between static observers is related to their clock rates; in a weak field, its logarithmic difference can be represented by a signed field integral along a directed path. With $\mathbf{g}$ defined by the inward-field sign convention,

$$\Delta \ln \delta\tau \simeq \frac{1}{c^2} \int_1^2 \mathbf{g} \cdot \mathrm{d}\boldsymbol{\ell}. \tag{13}$$

This is a weak-field relation and depends on the observer and sign conventions. If, in a specified setup, the signed field along the path can be represented by a time series and the path coordinate satisfies $\mathrm{d}\ell \simeq c\,\mathrm{d}u$, the integral takes the form of an acceleration average multiplied by L/c^2. The choice of $\mathcal{P}_+$ or $\mathcal{P}_-$ depends on how the path is oriented and

parameterized. The matching is therefore a model step requiring an explicit field-and-path map, not an identity for an arbitrary acceleration history.

This clock-rate application is logically separate from Wheeler–Feynman electrodynamics. An exact static clock-gradient formulation can be written using the lapse of a specified spacetime, but the distance parameter in that construction is not generally coordinate time. Derivatives with respect to it do not automatically inherit the physical time-domain meanings of jerk, snap, and higher derivatives used in Eq. (6). Neither the absorber theory nor the finite-window algebra alone establishes a universal strong-field clock law.

9 Conclusion

Wheeler–Feynman absorber theory is a physical proposal in which charged particles interact through a time-symmetric direct-action structure and the collective response of an absorber is invoked to account for radiation reaction and an effective retarded radiation description. Its account depends on absorber and boundary assumptions; time symmetry alone does not choose a radiation arrow.

The temporal propagator considered here is a different object: an exact finite-window average of a specified signal. Its future- and past-sampling versions have a precise even–odd decomposition, a conditional Taylor series, and a simple frequency response. Those properties can make it a useful mathematical comparison, but they do not turn it into an electromagnetic Green function. Keeping this distinction explicit allows the two subjects to illuminate one another structurally without conflating a window identity with the physical absorber mechanism.

References

[1] J. A. Wheeler and R. P. Feynman, "Interaction with the Absorber as the Mechanism of Radiation," *Reviews of Modern Physics* **17**, 157–181 (1945), doi:10.1103/RevModPhys.17.157.

[2] J. A. Wheeler and R. P. Feynman, "Classical Electrodynamics in Terms of Direct Interparticle Action," *Reviews of Modern Physics* **21**, 425–433 (1949), doi:10.1103/RevModPhys.21.425.

[3] P. C. W. Davies, "Is the Universe Transparent or Opaque?" *Journal of Physics A: General Physics* **5**, 1722–1737 (1972), doi:10.1088/0305-4470/5/12/012.